

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 1D nr. 5.3

$$(z + 2)^5$$

$\Rightarrow$ via Bin. Newton

$$(z + 2)^5 = \binom{5}{0}2^5 + \binom{5}{1}2^3z + \binom{5}{2}2^3 \cdot 2^2 + \binom{5}{3}2^2 \cdot 2^3 + \binom{5}{4}2 \cdot 2^4 + \binom{5}{5}2^5$$

$$= 1 \cdot 32 + 5 \cdot 16z + 10 \cdot 8z^2 + 10 \cdot 4z^3 + 5 \cdot 2z^4 + z^5$$

$$= 32 + 80z + 80z^2 + 40z^3 + 10z^4 + z^5$$

References